4-Channel Power Supply Controller Checklist

S/N:____________________ Date:__________ Initials:_______

☐ **1. Adjust Power Supplies**

A. Disconnect the following power connections prior to adjusting:
- Controller Filter Board: J1 & J2
- Digital Board: J2 & J4
- Regulator Board: J1 & J2

B. Power on the unit, ensure each power supply has the corresponding green LED lit.

C. Adjust the voltages as follows:

☐ +6.3V on rightmost Power Supply (Red lead on +V, Black lead on -V)
☐ -16.5V on middle Power Supply (Black lead on +V, Red lead on -V)
☐ +16.5V on leftmost Power Supply (Red lead on +V, Black lead on -V)

D. Turn off the Unit.

☐ **2. Install Jumpers and Fuses**

Controller Filter Board

☐ (2) White Jumpers
☐ (3) 3.15A Fuses

Digital Board

☐ (1) White Jumper
☐ (3) 3.15A Fuses

Regulator Board

☐ (55) White Jumpers[1] ☐ (2) 2.15A Fuses – [F1, F3]
☐ (1) 1A Fuse – [F2] ☐ (3) Blue Jumpers – [W1, W2, W3]

☐ **3. Reinstall Power Connections**

☐ **4. Perform Voltage Measurements (see page 2)**

☐ **5. Install SOM Board, Heat Sink, and Fan**
- Ensure Dip Switches are both pushed towards the left edge (SD Card Mode)
- Ensure SOM board green LED is lit [D3]
- Install the heat sink parallel to the Digital Board (air flow directed out of back)
- #4 flat washers may be required to ensure fan does not catch on heat sink

☐ **6. Install Burden Resistor Boards**

R1:__________ R2:__________ R3:__________ R4:__________

☐ **7. Install Compensation Boards**

X1:__________µF X2:__________Ω

☐ **8. Install SD Card into slot, ensure blue LED is lit. [D3]**

☐ **9. Unit Passed** **Initials__________**

Comments:____

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[1] Do not install: W16, W24, W28, W32. Install all vertical 3-prongs on the top two prongs. For horizontal 3-prong: Install W15, W23, W27, W31 on two left prongs; for the other 8 install on two right prongs.

Voltage Measurements

1. **Power on the Unit and Check the Following LEDs:**

   Note: If any are not lit, use a thermal imager to assess the board

   A. Regulator Board
      - (8) Orange [D52, D6, D8, D9, D12, D11, D16, D15]

   B. Digital Board
      - (3) Orange [DS1002, DS1001, DS1000]
      - (3) Green [D1, D5, D4]

2. **Measure the following on the Digital / Regulator Boards. Record values.**

   Note: U23 Tab is Universal Ground (G)

   1. TP2 → G. Value should be +6.2V ± 0.1V ___________V
   2. TP3 → G. Value should be +5.0V ± 0.1V ___________V
   3. TP4 → G. Value should be +1.8V ± 0.1V ___________V
   4. TP5 → G. Value should be +3.3V ± 0.1V ___________V
   5. TP6 → G. Value should be +1.0V ± 0.1V ___________V
   6. TP7 → G. Value should be +1.2V ± 0.1V ___________V
   7. TP9 → G. Value should be +1.8V ± 0.1V ___________V
   8. U30 pin 1 → G. Value should be -16.5V ± 0.1V ___________V
   9. U30 pin 4 → G. Value should be -15.0V ± 0.1V ___________V
   10. U33 pin 4 → G. Value should be +10V ± 0.1V ___________V
   11. U34 pin 4 → G. Value should be +5.3V ± 0.1V ___________V
   12. U29 pin 3 → G. Value should be +15.0V ± 0.1V ___________V
   13. U29 pin 1 → G. Value should be +16.5V ± 0.1V ___________V
   14. U35 pin 3 → G. Value should be +2.5V ± 0.1V ___________V
   15. U32 pin 5 → G. Value should be -5.25V ± 0.1V ___________V
   16. U31 pin 5 → G. Value should be -9.85V ± 0.15V ___________V
   17. U1003 pin 4 → G. Value should be +3.3V ± 0.1V ___________V

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   1. U1 pin 3 → G. Value should be +5.0V ± 0.1V ___________V
   2. U7 pin 3 → G. Value should be +5.0V ± 0.1V ___________V
   3. U53 pin 1 → G. Value should be +16.5V ± 0.1V ___________V
   4. U53 pin 3 → G. Value should be +15.0V ± 0.2V ___________V
   5. U54 pin 1 → G. Value should be -16.5V ± 0.1 V ___________V
   6. U54 pin 5 → G. Value should be -15.0V ± 0.2V ___________V
   7. U51 pin 3 → G. Value should be +15.0V ± 0.2V ___________V
   8. U52 pin 5 → G. Value should be -15.0V ± 0.2V ___________V
   9. U43 pin 3 → G. Value should be +15.0V ± 0.2V ___________V
   10. U50 pin 5 → G. Value should be -15.0V ± 0.2V ___________V